

Optical Heart Rate Sensor Final Project

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Abstract

This project report details the design, implementation, and testing of a four-stage non-invasive optical heart rate sensor utilizing photoplethysmography (PPG). The objective was to engineer an analog processing chain capable of isolating and amplifying a human pulse from a noisy optical signal and converting it into a digital output and physical indicator. The system consists of an infrared emitter and phototransistor stage to detect a human pulse, followed by cascaded active high-pass and low-pass filters providing a total system gain of 54.87 V/V to filter noise and amplify the heartbeat. A final comparator stage was used with a 99 mV hysteresis window to drive a physical LED indicator and provide a clean digital square wave for heart rate calculations.

Experimental results via Frequency Response Analysis (FRA) confirmed a sufficient bandpass response with a bandwidth from 0.370 to 5.03 Hz, effectively isolating the target 40 – 200 BPM range. Functional testing demonstrated high accuracy with a measured resting heart rate of 90 BPM, creating only a 3.33% error compared to an Apple Watch. While component tolerances of $\pm 20\%$ in electrolytic capacitors led to slight deviations from theoretical cutoff frequencies, the system remained fully functional and stable. This project successfully demonstrates active analog filtering used in medical practices.

List of Parts: IR204 Infrared Emitter, PT204-6B Phototransistor, LM324N Operational Amplifiers, LM339 Comparator, NMOS RFD3055LE, resistors, capacitors, green LED, Analog Discovery 2.

Introduction

The objective of this project is to design and test a non-invasive optical heart rate sensor that can detect a wide range of pulses from a user and display it in the form of a digital oscilloscope waveform and a physical flashing LED.

Monitoring heart rate is a vital task in health diagnosis that can help professionals understand their patients' issues and provide quality care. However, capturing a clean pulse signal electronically presents several challenges such as noise, movement, and non-idealities of electrical components. The primary method used to detect a heartbeat is photoplethysmography (PPG), which detects changes in blood volume in the capillaries via light absorption. However, the raw signal from an optical sensor like an infrared emitter is often buried under a large DC offset and significant noise from ambient light and electrical interference. Without precise analog

processing, these small millivolt changes cannot be reliably converted to a digital output and beats per minute (BPM) value.

The solution for this project is comprised of 4 separate stages that form an analog processing chain to isolate and amplify a human pulse. Stage 1 is the IR emitter and receiver circuit in which an infrared emitting diode is placed on one side of the user's finger and a phototransistor on the other. The infrared LED and phototransistor act as the transducer to convert blood flow changes into a voltage signal output that is fed into Stage 2. The second stage and third stage is a series of active filters (high-pass and low-pass) which are specifically tuned to the frequency of a human heartbeat (40- 200 BPM) which also provides significant voltage gain. The amplified signal is the input to the last stage which is comprised of an analog-to-digital interface created by using a comparator with a hysteresis loop to drive a green LED. So, the LED will blink at the same rate as the user's heartbeat.

Theory

Diving into each stage of the solution:

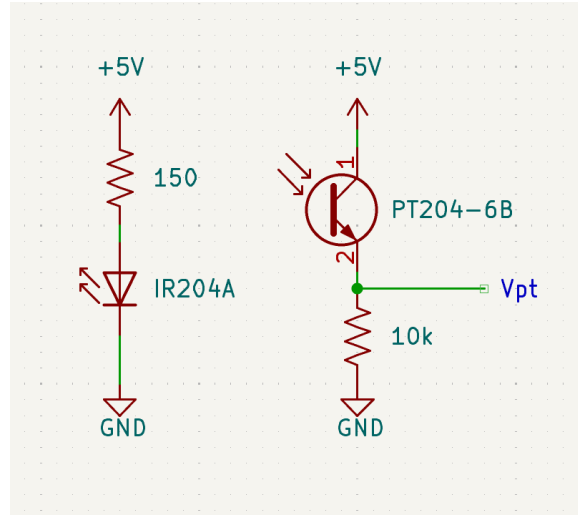


Figure 1: Stage 1 Optical Sensor Circuit Schematic

Stage 1 – Optical Sensor (Transducer)

- Purpose: To convert physical changes in blood volume into an electrical voltage signal using photoplethysmography (PPG).

- Operation: An IR LED (IR204) emits infrared light through the finger. As the heart beats, blood volume in the capillaries change, which increases and decreases the amount of light reaching the phototransistor (PT204-6B).
- Circuit Design: The phototransistor is biased in an emitter-follower configuration. The current I_{PT} flowing through the transistor is proportional to the light received, creating an output voltage $V_{PT} = I_{PT} \times R_E$. R_E is calculated to be a value that makes V_{PT} have a large enough DC voltage ($>1V$) with a small AC peak to peak value (10-100mV) that is the user's heartbeat.

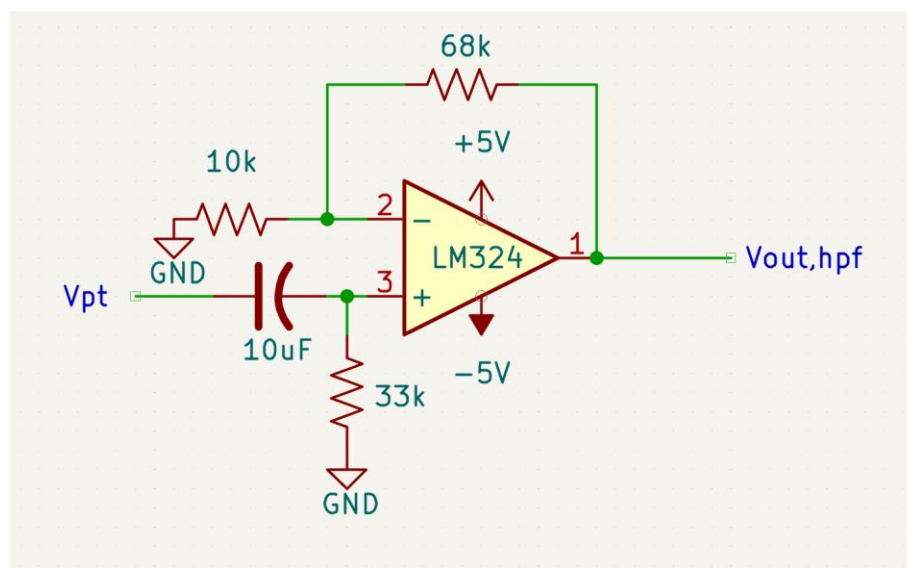


Figure 2: Stage 2 High-Pass Filter (HPF) Circuit Schematic

Stage 2: High-Pass Filter (HPF)

- Purpose: To block the large DC offset from the sensor and allow only the pulsating AC signal (the heartbeat) to pass.
- Operation: This is an active non-inverting high-pass filter using the LM324 op-amp. It provides a sharp cutoff for low frequencies (like finger movement or slow DC drifts) while amplifying the signal.
- Circuit Design: The voltage V_{PT} is fed from the optical sensor stage into a high-pass filter comprised of a capacitor and resistor in parallel. The values of these components determine the range of frequency that will be let through the filter, and which will be stopped. For frequencies that pass through the filter, a non-inverting op-amp is used to amplitude the AC signal for stage 4. The gain for this op-amp can be customized based on the input resistor R_{IN} , and feedback resistor R_F .

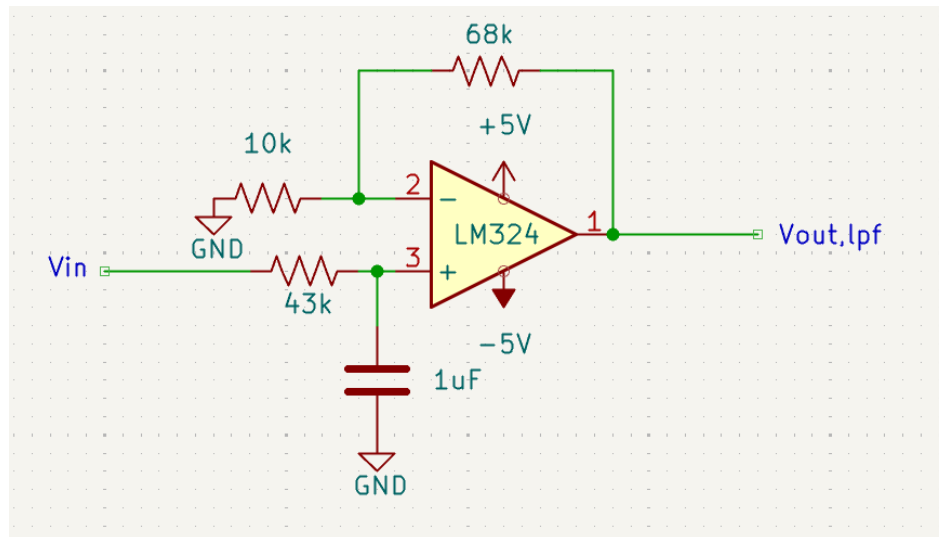


Figure 3: Stage 3 Low-Pass Filter (LPF) Circuit Schematic

Stage 3: Low-Pass Filter (LPF)

- Purpose: To remove high-frequency noise (such as 60 Hz interference from room lights) and smooth the heartbeat waveform.
- Operation: This is an active non-inverting low-pass filter. By cascading this with the HPF, a bandpass filter is made which isolates frequencies between roughly 0.5 to 4.25 Hz, the range of a human heartbeat.
- Circuit Design: Continuing through the circuit, the output of the high-pass filter is fed into a low-pass filter which is a flipped component version of the HPF using a capacitor and resistor. Different values are needed for these components to determine the cutoff frequency for this filter which only lets lower frequencies through and cuts off frequencies from other sources like ambient light so only the user's heart rate is passes through. Another non-inverting op-amp stage is used after the filter to continue to amplify the AC heartbeat signal to between 2 and 4 V_{p-p} .

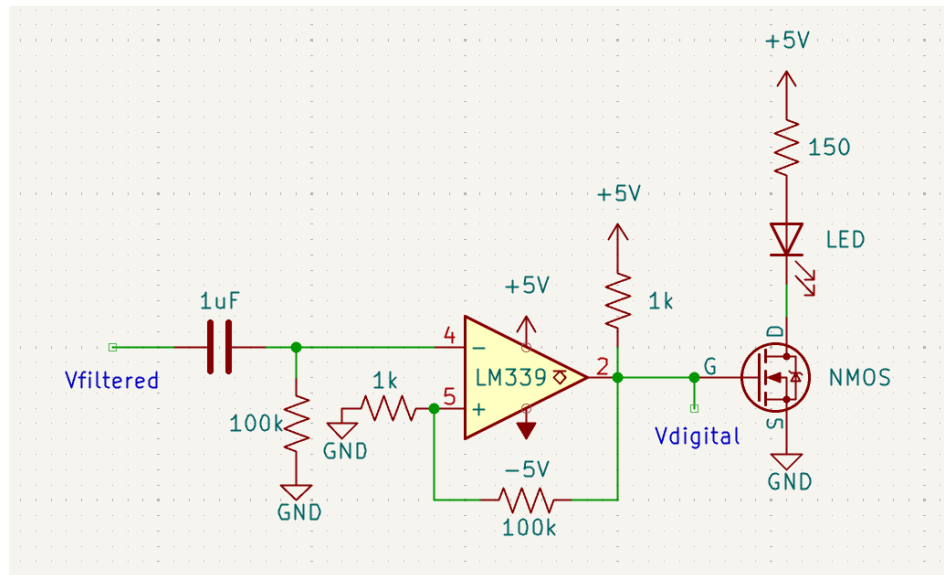


Figure 4: Stage 4 Comparator and LED Display Circuit Schematic

Stage 4: Analog to Digital (Comparator) + LED Display

- Purpose: To convert the filtered analog heartbeat into a clean square wave that can trigger a digital indicator (LED).
- Operation: The LM339 comparator compares the analog signal to a reference voltage and depending on the input voltage will either +5V or -5V that can be used to toggle the LED indicator.
- Hysteresis: Using a 100 k Ω feedback resistor creates a hysteresis loop. This prevents multiple triggers from a single heartbeat by requiring the signal to pass two different thresholds (high and low) before switching the output state.
- Circuit Design: The final stage of the system utilizes an LM339 quad comparator to interface the analog heartbeat with a digital display. The output of the low-pass filter is AC-coupled through a capacitor to the inverting input of the comparator, centered around a ground reference using a 100 k Ω resistor. To provide noise immunity and prevent multiple triggers during transitions, a 100 k Ω feedback resistor is connected to the non-inverting input to create a hysteresis window. This digital square wave drives the gate of an RFD3055LE NMOS, which acts as a switch that toggles a green LED.

Equation Derivations:

Cutoff Frequency for High-Pass and Low-Pass Filters:

$$f_{\text{cutoff}} = \frac{1}{2\pi RC}$$

Both the High-Pass and Low-Pass filters rely on a passive RC network at the non-inverting input of the op-amp.

Using voltage division:

- For a Low-Pass Filter, the output of the RC stage is taken across the capacitor:

$$V_{\text{out}} = V_{\text{in}} \times \frac{Z_C}{R+Z_C}, Z_C = \frac{1}{j\omega C}$$

- For a High-Pass Filter, the output is taken across the resistor:

$$V_{\text{out}} = V_{\text{in}} \times \frac{R}{R+Z_C}, Z_C = \frac{1}{j\omega C}$$

For both configurations, the "cutoff" occurs when the magnitude of the resistance (R) equals the magnitude of the capacitive reactance (Z_C):

$$R = \frac{1}{\omega C}$$

Rearrange and solve for ω_c :

$$\omega_c = \frac{1}{RC}$$

Since $\omega = 2\pi f$:

$$f_c = \frac{1}{2\pi RC}$$

Non-Inverting Operational Amplifier Gain:

$$\text{Gain } (A_v) = 1 + \frac{R_f}{R_1}$$

Assume ideal operational amplifier behavior:

- 1) No current flows into the input terminals ($I_+ = I_- = 0$).
- 2) The voltage at the two input terminals is equal ($V_+ = V_-$).

Apply Kirchhoff's Current Law (KCL) at the inverting terminal (V_-). The current flowing from the output through R_f must equal the current flowing through R_1 to ground:

$$\frac{V_{out} - V_-}{R_f} = \frac{V_- - 0}{R_1}$$

Since $V_+ = V_{in}$ (the signal from our RC network) and $V_+ = V_-$, substitute V_{in} into the equation:

$$\frac{V_{out} - V_{in}}{R_f} = \frac{V_{in}}{R_1}$$

Solve for V_{out} :

$$\frac{V_{out}}{R_f} = \frac{V_{in}}{R_1} + \frac{V_{in}}{R_f}$$

$$\frac{V_{out}}{R_f} = V_{in} \left(\frac{1}{R_1} + \frac{1}{R_f} \right)$$

$$V_{out} = V_{in} \left(\frac{R_f}{R_1} + \frac{R_f}{R_f} \right)$$

Solve for Gain (A_v) = $\frac{V_{out}}{V_{in}}$:

$$\text{Gain } (A_v) = \frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_1}$$

Design / Calculations

Stage 1: Optical Sensor Biasing

The IR204 infrared LED requires a specific amount of forward current I_F to produce enough light intensity to penetrate the finger tissue.

- Design Choice: A 150 Ω resistor was selected to limit the current (R_D).
- Calculation: Using Kirchhoff's Voltage Law (KVL) around the emitter loop, and assuming a standard forward voltage V_F of approximately 1.2 to 1.5 V (at 20 mA) for an IR LED:

$$I_F = \frac{V_{CC} - V_F}{R_D} = \frac{5 \text{ V} - 1.3 \text{ V}}{150 \Omega} \approx 24.7 \text{ mA}$$

- Justification: This current level is high enough to ensure the light is bright enough to be picked up by the phototransistor on the opposite side of the finger, while staying safely below the maximum rated current for the device which is 100 mA.

The emitter resistor, R_E , for the PT204-6B was chosen to place the transistor in the forward active region, which is necessary for linear sensing.

- Design Choice: After testing, a 10.3 k Ω resistance was used.

- Calculation: To keep the transistor active, the collector-emitter voltage V_{CE} must be greater than the saturation voltage $V_{CE(SAT)} = 0.4 \text{ V}$.
- Experimentally measured V_{PT} :

$$V_{PT} = i_{PT} \times R_E = 0.138 \text{ mA} * 10.3 \text{ k}\Omega = 1.42 \text{ V}$$

This provided a stable DC offset while allowing for an AC heartbeat ripple of $\approx 83 \text{ mV}_{p-p}$.
- Verification: $V_{CE} = 5 \text{ V} - 1.42 \text{ V} = 3.58 \text{ V}$. Since $3.58 \text{ V} > 0.4 \text{ V}$, the sensor is properly biased for linear heart rate detection.

Stage 2/3: Filters and Gain Calculations

The filters were designed for a human heart rate range of 40 to 200 BPM, which corresponds to 0.67 Hz to 3.33 Hz.

High Pass Filter:

- Design Choice: $R = 33 \text{ k}\Omega$ and $C = 10 \text{ }\mu\text{F}$
- Cutoff Frequency Calculation:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(33 \text{ k}\Omega)(10 \text{ }\mu\text{F})} = 0.482 \text{ Hz}$$

- Justification: The high pass filter cutoff needs to be less than 0.67 Hz to give a buffer to low heart rates. Using $R = 33 \text{ k}\Omega$ and $C = 10 \text{ }\mu\text{F}$ gives the filter a cutoff of 0.482 Hz which is lower than the 0.67 Hz parameter indicating good values that allow the filter to output just the heartbeat signal and not the DC offset that V_{PT} has.

Low Pass Filter:

- Design Choice: $R = 43 \text{ k}\Omega$ ($33 \text{ k}\Omega + 10 \text{ k}\Omega$ in series) and $C = 1 \text{ }\mu\text{F}$
- Cutoff Frequency Calculation:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(43 \text{ k}\Omega)(1 \text{ }\mu\text{F})} = 3.701 \text{ Hz}$$

- Justification: The low pass filter cutoff needs to be higher than 3.33 Hz to give a buffer to high heart rates. With these component values the filter has a cutoff of 3.701 Hz which is higher than design parameter which allows the filter to output just the heartbeat signal and block high frequency fluctuations like from ambient light.

High Pass Filter Op-Amp:

- Design Choice: $R_F = 68 \text{ k}\Omega$ and $R_1 = 10 \text{ k}\Omega$
- Gain Calculation:

$$\text{Gain } (A_{v1}) = 1 + \frac{R_f}{R_1} = 1 + \frac{68 \text{ k}\Omega}{10 \text{ k}\Omega} = 7.8$$

- Justification: This stage provides the initial amplification needed to lift the millivolt-level signal from the sensor. By using $68 \text{ k}\Omega$ instead of a higher value, the stage provides a significant gain of 7.8 while maintaining a voltage that is sufficient to be amplified again in the low pass stage and prevents the signal from hitting the $\pm 5\text{V}$ power rails and clip the output.

Low Pass Filter Op-Amp:

- Design Choice: $R_F = 68 \text{ k}\Omega$ and $R_1 = 10 \text{ k}\Omega$
- Gain Calculation:

$$\text{Gain } (A_{v2}) = 1 + \frac{R_f}{R_1} = 1 + \frac{68 \text{ k}\Omega}{10 \text{ k}\Omega} = 7.8$$

- Justification: Since the stages are cascaded, this second stage of amplification brings the signal to its final analog amplitude. Using identical resistor values simplifies the design and ensures the total gain is distributed evenly across the filter chain.

Total Gain:

- Calculation:

$$\text{Total Gain } (A_v) = A_{v1} \times A_{v2} = 7.8 \times 7.8 = 60.84$$

- Justification: With a total gain of approximately 60.8, a typical heartbeat signal of $50 - 80 \text{ mV}_{\text{p-p}}$ from Stage 1 is amplified to roughly $3 - 5 \text{ V}_{\text{p-p}}$. This creates a robust analog waveform that is large enough to reliably trip the comparator thresholds in Stage 4 without causing the op-amps to saturate and clip the signal.

Stage 4: Comparator and LED Display

- Design Choice: $R_1 = 1 \text{ k}\Omega$, $R_2 = 100 \text{ k}\Omega$ where R_2 is the feedback resistor.
- Hysteresis Calculation: The hysteresis voltage V_{hys} is the difference between the upper and lower trip thresholds. It is calculated based on the ratio of the series and feedback resistors and the output voltage swing (10 V total for $\pm 5\text{V}$ rails):

$$V_{hys} \approx V_{CC} - V_{SS} \times \frac{R_1}{R_1 + R_2} = (5V + 5V) \times \frac{1k\Omega}{1k\Omega + 100k\Omega} \approx 99mV$$

- Justification: This 99 mV window effectively shields the digital output from low-level analog noise. It ensures that the comparator only toggles when a clear heartbeat pulse is detected, preventing the LED from flickering or due to small fluctuations in the filtered signal.

LED Driver Circuit:

- Design Choice: $R_L = 150\ \Omega$ and an RFD3055LE NMOS.
- Current Calculation:

The standard forward voltage $V_F = 2.1\ V$ for a green LED:

$$I_L = \frac{V_{CC} - V_F}{R_L} = \frac{5V - 2.1V}{150\ \Omega} = 19.3\ mA$$

- Justification: This current level provides high brightness for the physical visual indicator while remaining well within the 30 mA maximum rating of a standard green LED. Using a NMOS as a switch ensures that the comparator output is not directly affected by the LED current.

Results

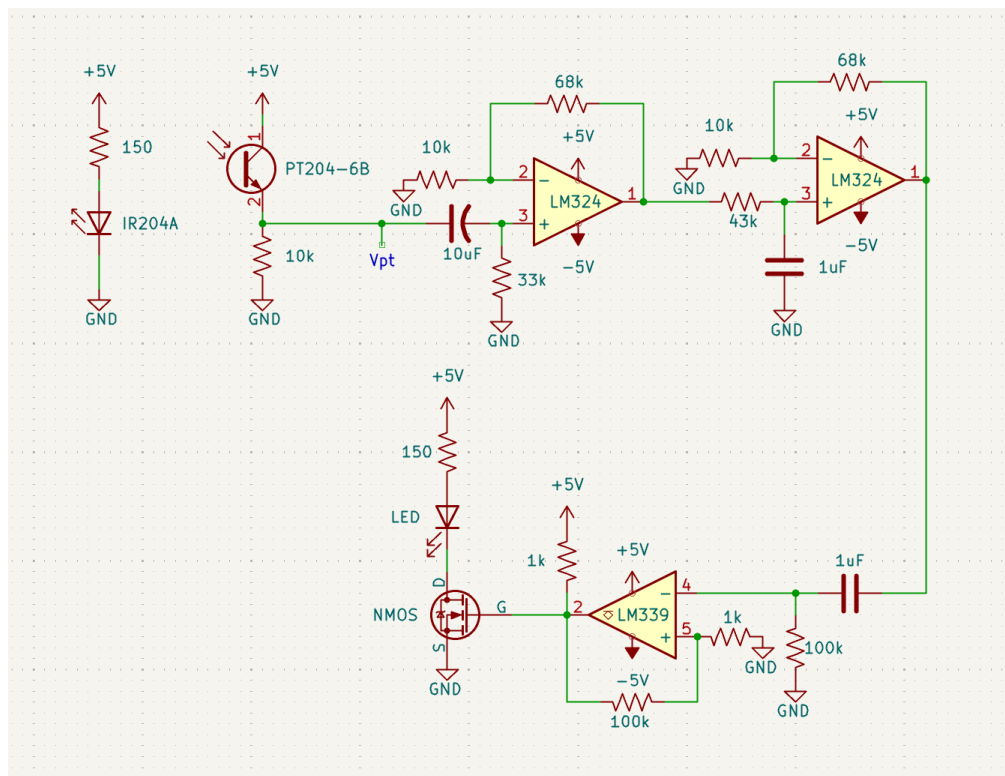


Figure 5: Full Heartbeat Sensor Circuit Schematic

Subsystem 1: Optical Sensor Output

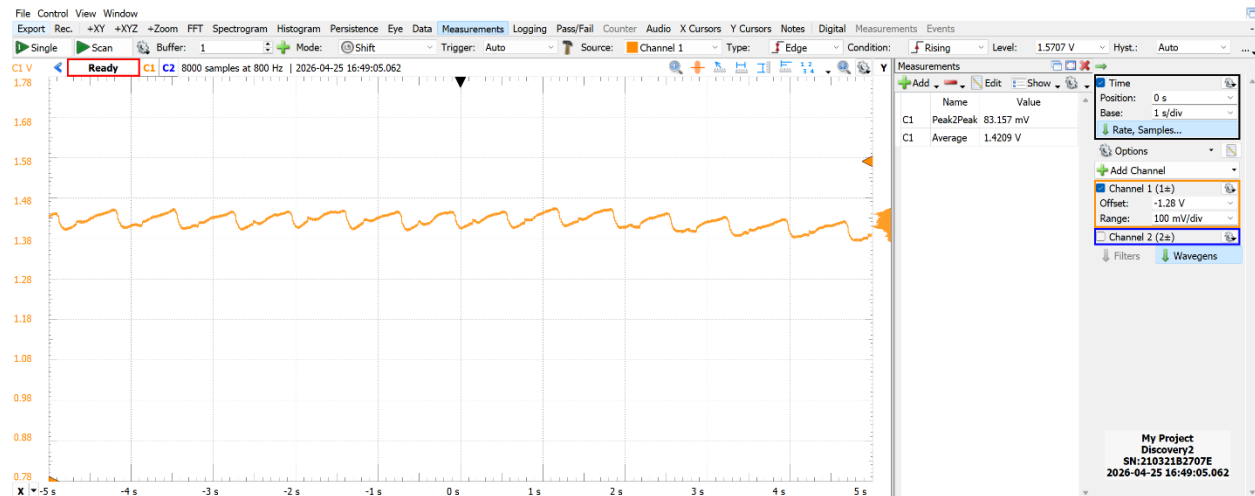


Figure 6: V_{PT} Voltage Output from Optical Sensor

Figure 6 shows the raw signal output by the phototransistor with a finger in place. The signal has a measured average (DC) value of 1.42 V. Within this DC offset, a periodic AC ripple of approximately 83.16 mV p-p is clearly visible, representing the change in light absorption caused by the user's heartbeat. This confirms that Stage 1 is properly biased in the forward active region.

Subsystem 2/3: High and Low Pass Filter Frequency Response Analysis

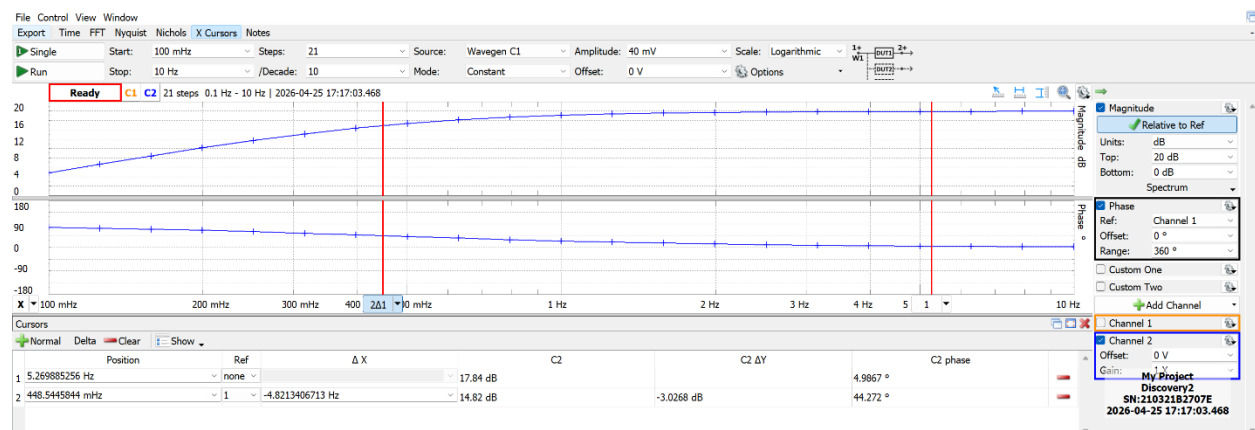


Figure 7: Frequency Response Analysis of Active High-Pass Filter (Logarithmic Gain)

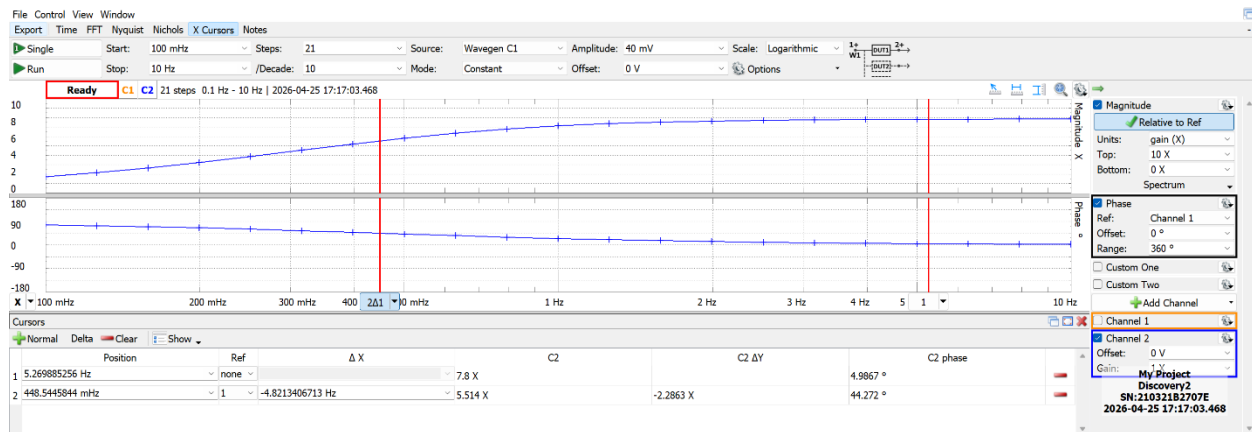


Figure 8: Frequency Response Analysis of Active High-Pass Filter (Linear Gain)

High-Pass Filter (HPF) Results

The High-Pass Filter was designed with a theoretical cutoff of 0.482 Hz and a gain of 7.8 V/V.

- Figure 7 Logarithmic Plot Analysis:
 - Magnitude (dB): The plot shows a clear +20dB/decade slope at frequencies below the cutoff, which is characteristic of a first-order filter.
 - Cutoff: At 448 mHz, the gain drops by -3 dB from its peak, which is close to the designed cutoff frequency.
- Figure 8 Linear Plot Analysis:
 - Magnitude (Gain X): In the passband (frequencies above 1 Hz), the gain stabilizes at approximately 7.8, confirming the accuracy of the non-inverting op-amp gain equation.
 - Phase Shift: At the measured cutoff frequency, the phase is 44.27°, providing verification of the RC network's behavior which is 45°

High-Pass Filter % Error Calculations

Cutoff Frequency (f_L):

$$\% \text{ Error} = \frac{|0.448 \text{ Hz} - 0.482 \text{ Hz}|}{0.482 \text{ Hz}} \times 100 = 7.05\%$$

Gain (A_{v1}):

$$\% \text{ Error} = \frac{|7.8 - 7.8|}{7.8} \times 100 \approx 0\%$$

Phase:

$$\% \text{ Error} = \frac{|44.27^\circ - 45^\circ|}{45^\circ} \times 100 = 1.62\%$$

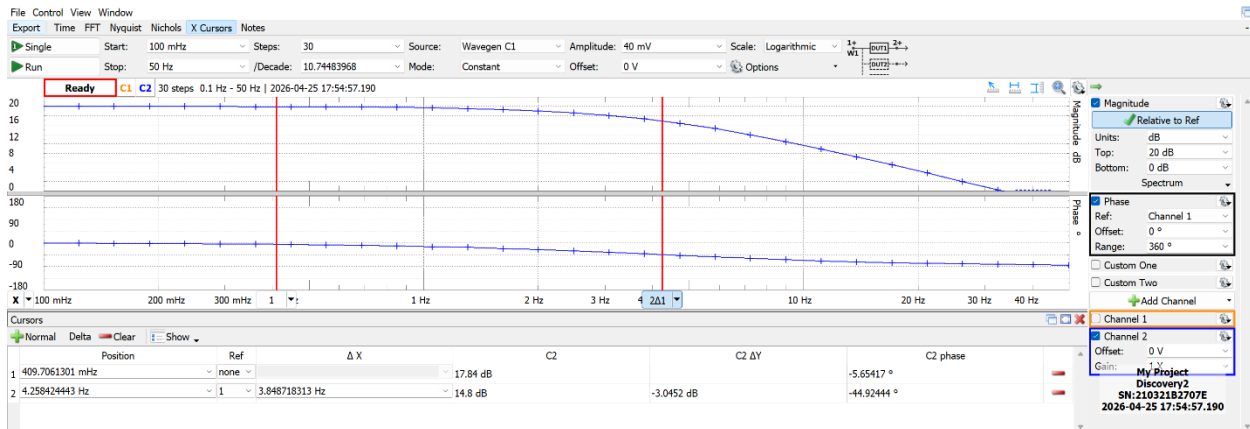


Figure 9: Frequency Response Analysis of Active Low-Pass Filter (Logarithmic Gain)

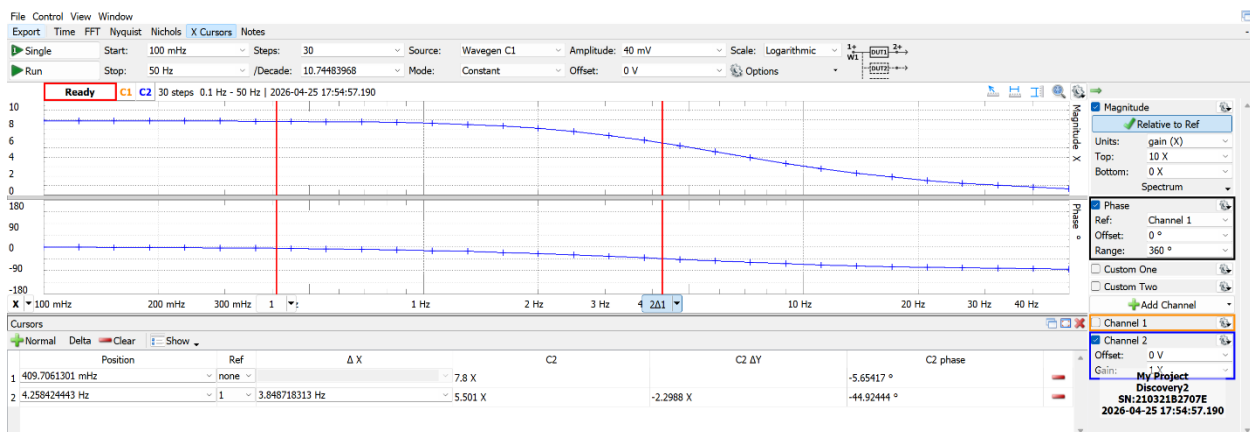


Figure 10: Frequency Response Analysis of Active Low-Pass Filter (Linear Gain)

Low-Pass Filter (LPF) Results

The Low-Pass Filter had a theoretical cutoff of 3.701 Hz.

- Figure 9 Logarithmic Plot Analysis:
 - Magnitude (dB): The plot demonstrates a -20dB/decade attenuation for frequencies higher than the cutoff, effectively suppressing high-frequency interference like 60 Hz ambient light.
 - Cutoff: Cursors identify the -3 dB point at 4.258 Hz. While higher than the 3.701 Hz design, this remains within the acceptable range for heart rate detection.
- Figure 10 Linear Plot Analysis:
 - Magnitude (Gain X): The low-frequency gain is measured at 7.8, matching the HPF stage as the resistors of the non-inverting op-amps are the same.
 - Phase Shift: The phase at the 4.258 Hz cutoff was 44.92° which confirms a successful first-order low-pass implementation.

Low-Pass Filter % Error Calculations

Cutoff Frequency (f_H):

$$\% \text{ Error} = \frac{|4.258 \text{ Hz} - 3.701 \text{ Hz}|}{3.701 \text{ Hz}} \times 100 = 15.05\%$$

Gain (A_{v2}):

$$\% \text{ Error} = \frac{|7.8 - 7.8|}{7.8} \times 100 \approx 0\%$$

Phase:

$$\% \text{ Error} = \frac{|44.92^\circ - 45^\circ|}{45^\circ} \times 100 = 0.18\%$$

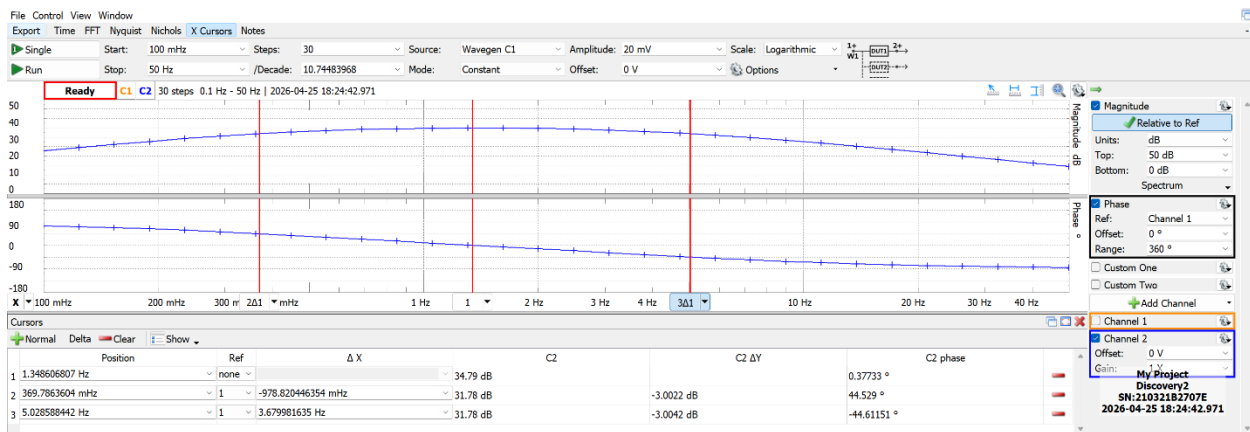


Figure 11: Frequency Response Analysis of Band-Pass Filter (HPF + LPF) (Logarithmic Gain)

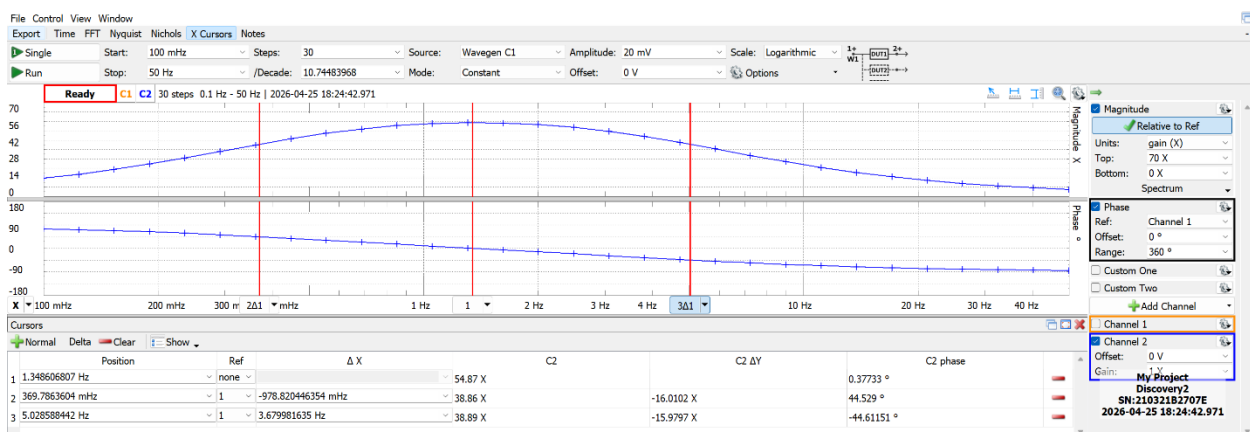


Figure 12: Frequency Response Analysis of Band-Pass Filter (HPF + LPF) (Linear Gain)

Band-Pass Filter (BPF) Results

The BPF results represent the cascaded performance of Stage 2 and Stage 3.

- Figure 11 Logarithmic Plot Analysis:
 - Bandwidth: The plot forms a "hump" shape. The -3 dB points on either side of the peak defines the system bandwidth. It was measured from 370 mHz to 5.03 Hz. This range is ideal for capturing heart rates from 22 BPM to 300 BPM.
 - Peak Gain: The peak gain is measured at 34.79 dB.
- Figure 12 Linear Plot Analysis:
 - Total System Gain: The peak linear gain is 54.87. This is slightly lower than the theoretical 60.84 (7.8×7.8) due to the overlapping attenuation of the two filters at the center frequency.
 - Center Frequency: The peak is centered at 1.35 Hz (81 BPM) which is the optimal sensitivity zone for a resting human pulse.

Band-Pass Filter % Error Calculations

Low Cutoff Frequency (f_L):

$$\% \text{ Error} = \frac{|0.370 \text{ Hz} - 0.482 \text{ Hz}|}{0.482 \text{ Hz}} \times 100 = 23.24\%$$

High Cutoff Frequency (f_H):

$$\% \text{ Error} = \frac{|5.03 \text{ Hz} - 3.701 \text{ Hz}|}{3.701 \text{ Hz}} \times 100 = 35.91\%$$

Gain (A_v):

$$\% \text{ Error} = \frac{|54.87 - 60.84|}{60.84} \times 100 = 9.81\%$$

The measured lower cutoff of 0.370 Hz and upper cutoff 5.03 Hz show the highest percent errors. These discrepancies are largely expected due to component tolerances. The 10 μF and 1 μF electrolytic capacitors have a tolerance of $\pm 20\%$. Additionally, when stages are cascaded, the interaction between the high-pass and low-pass curves slightly shifts the -3 dB points compared to the individual stage measurements. Despite the error, the filter still successfully isolates the target range of 22 to 301 BPM satisfying the design requirement for a heart rate sensor from 40 to 200 BPM.

The measured peak gain of 54.87 is 9.81% lower than the theoretical 60.84. This variation is due to the non-ideal behavior of the LM324 op-amp and the overlap of the two filters. In a bandpass configuration, the gain at the center frequency is slightly attenuated by both

the high-pass and low-pass slopes, preventing the system from reaching the full mathematical maximum of $A_{v1} \times A_{v2}$. Even though the total gain is less than theoretical, 54.87 is still sufficient to provide a robust signal for the comparator stage without saturating the $\pm 5V$ rails.

The phase at the center frequency, 1.348 Hz, was measured at 0.377° . Theoretically, the center frequency of a bandpass filter is the point where the phase shift is exactly 0° . A measured value so close to zero confirms that the 1.348 Hz peak is the true resonant center of the active filter chain. The negligible difference is likely due to minor phase lag within the op-amp stages.

Subsystem 4: Comparator + LED Display

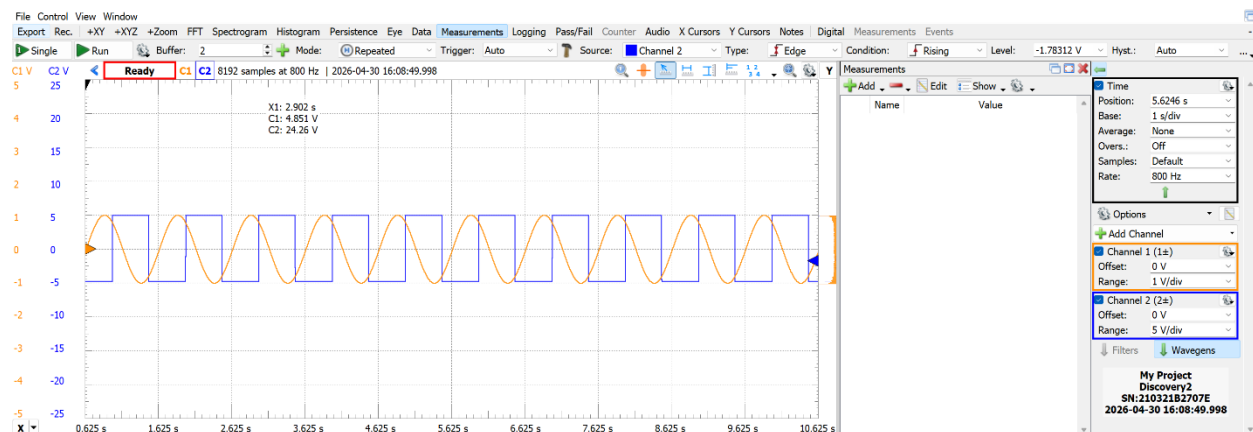


Figure 13: V_{hys} Comparator Verification Using 1 Hz Input Sine Wave

Figure 13 illustrates a Schmitt trigger circuit (a comparator with 99 mV of hysteresis) converting a 1 Hz sine wave (orange) into a clean, 50% duty cycle square wave (blue). At a frequency of 1 Hz, each full cycle takes exactly one second which is the expected output indicating a working comparator. The output goes from -5V to 5V, meaning the comparator is using its rails correctly.

Fully Integrated Project:

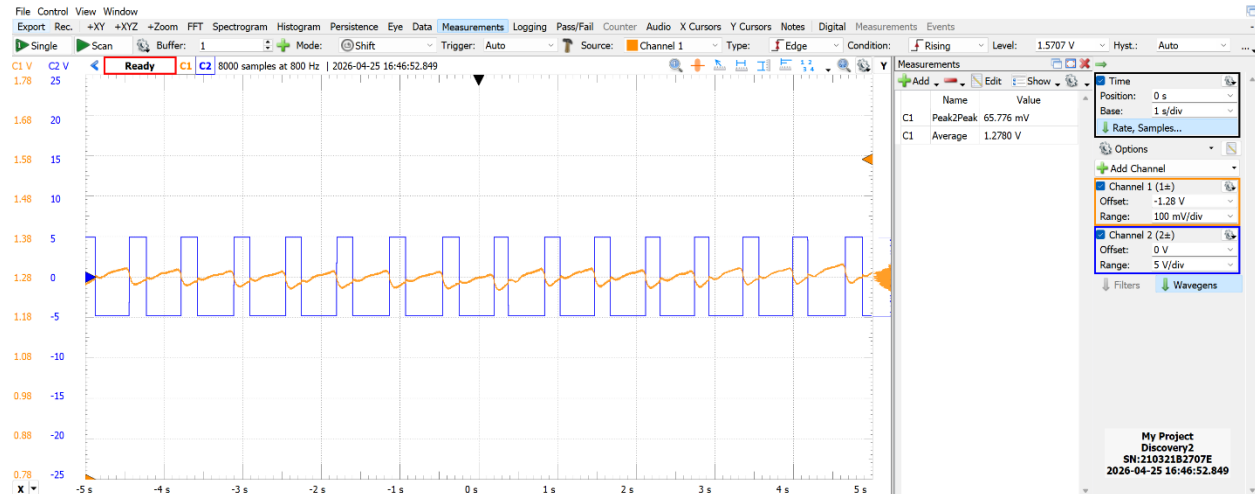


Figure 14: Final Synchronized Analog Pulse and Digital Output

Figure 14 is a dual-channel oscilloscope plot that displays the analog heartbeat against the digital square wave produced by the LM339 comparator. The digital output (blue pulses) triggers exactly on the falling edges of the analog heartbeat pulses (orange). This confirms that the threshold levels are correctly set. The 100 kΩ feedback resistor successfully implemented a 99 mV hysteresis window. The lack of multiple digital triggers per single analog peak proves that the design effectively ignores low-level noise. The digital output swings the full range from -5 V to +5 V, confirming the 1 kΩ pull-up resistor is functioning correctly with the open-collector output of the LM339. The digital signal successfully drives the NMOS switch to toggle the green LED at the same rate as the user's heartbeat, providing a clear physical demonstration of functionality.

BPM Accuracy:

From the oscilloscope plot over a period of 10 seconds, there are 15 pulses.

$$\text{Heart Rate} = \frac{\text{Pulses}}{\text{Period}} \times 60 = \frac{15}{10} \times 60 = 90 \text{ BPM}$$

Heart Rate % Error Calculations:

During this measurement, the user had a heart rate of 93 BPM according to an Apple Watch.

$$\% \text{ Error} = \frac{|90 - 93|}{93} \times 100 = 3.33\%$$

Given the 90 BPM example above was also a full implementation of a working heart rate sensor, more experimenting is needed to show that the circuit has correct output a human pulse for a wide range of beats per minute. The following Figure 14 shows the digital output for an elevated heart range of around 140-145 BPM.

146 BPM (Apple Watch Measurement) -

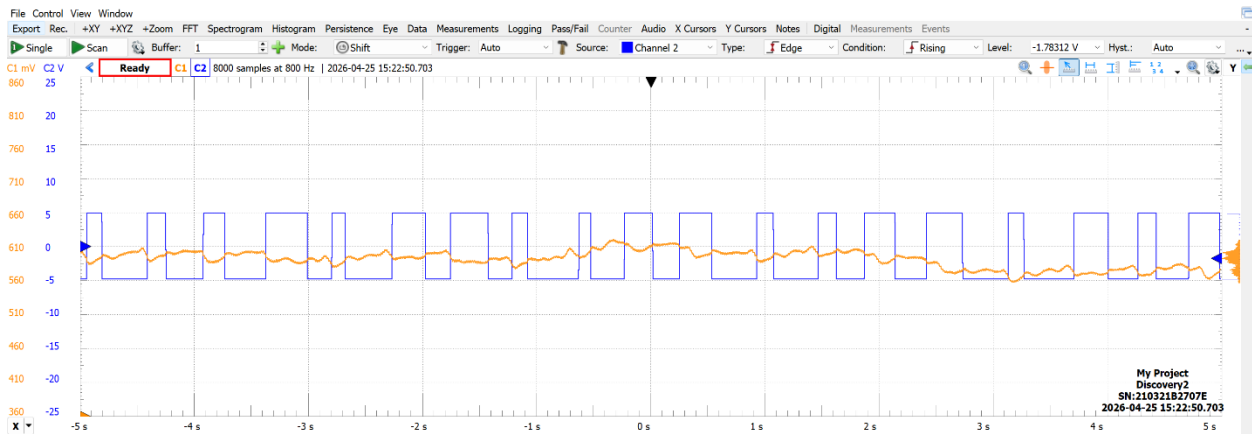


Figure 14: Heart Rate Output Measurement (146 BPM)

In Figure 14, the input wave is less smooth than in Figure 13. This is due to having an elevated heart rate (130+ BPM) making it hard to keep a finger still on the IR sensor causing more noise than expected experimentally. Still the circuit performed as expected as the comparator would go high on the falling edge of the input.

BPM Accuracy:

From the oscilloscope plot over a period of 4 seconds, there are 9 pulses.

$$\text{Heart Rate} = \frac{\text{Pulses}}{\text{Period}} \times 60 = \frac{9}{4} \times 60 = 135 \text{ BPM}$$

Heart Rate % Error Calculations:

During this measurement, the user had a heart rate of 146 BPM according to an Apple Watch.

$$\% \text{ Error} = \frac{|135 - 146|}{146} \times 100 = 7.53\%$$

There exists some variation as smartwatches often use a "rolling average" over 15–30 seconds, whereas this project is looking at shorter period only a few seconds. Heart rate drops rapidly once you stop running so after stopping and using the sensor, the user's heart rate is expected to start descending. The Apple Watch uses green light (reflective PPG), while this circuit uses infrared (transmissive PPG), which can lead to slight timing variations in peak detection.

Conclusion

The design and implementation of the four-stage optical heart rate monitor were successful, as the system effectively isolated, amplified, and digitized a human pulse. The integrated high and low pass filters successfully isolated the target frequency range for human heart rates (40 – 200 BPM) with a measured peak gain of 54.87. Furthermore, the comparator stage provided a clean digital square wave that accurately triggered the LED indicator in synchronization with the user's heartbeat.

The Frequency Response Analysis (FRA) confirmed that the system operated as a robust bandpass filter with cutoffs at 0.370 Hz and 5.03 Hz. While these values differed from the theoretical design of 0.482 Hz and 3.701 Hz, the primary sources of error are attributed to component tolerances, specifically the $\pm 20\%$ variation common in electrolytic capacitors such as the 10 μF and 1 μF used in the filters, and the cascading effects of active filter stages.

The system demonstrated high accuracy during experimental testing, with a measured heart rate of 90 BPM compared to a reference value of 93 BPM from an Apple Watch, resulting in a low percent error of only 3.33%. Although noise increased at higher heart rates due to finger movement sensitivity, the hysteresis window of 99 mV successfully prevented false triggers and ensured a reliable digital output. Overall, the project met all technical specifications and successfully demonstrated the principles of analog signal processing for medical applications.

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